Catheterization and Cardiovascular Interventions 70:999–1007 (2007)

PEDIATRIC AND CONGENITAL HEART DISEASE

Original Studies

Patent Foramen Ovale and Unexplained Ischemic Cerebrovascular Events in Children

Mirjana Perkovič Benedik, MD, MSc, **Marjan Zaletel,** MD, PhD, **Nuška Pečarič Meglič,** MD, **and Tomaž Podnar,*** MD, PhD

Objectives: To consider the role of patent foramen ovale (PFO) in ischemic cerebrovascular event of unknown cause in children. **Background:** Data regarding the possibility of paradoxical embolism in unexplained ischemic cerebrovascular event in children are lacking. **Methods:** Between January 2005 and March 2007, all consecutive children evaluated due to ischemic cerebrovascular event were included in the retrospective study. In addition to the standard diagnostic protocol, a contrast transcranial Doppler (TCD) with Valsalva maneuver (VM) was performed in patients with unexplained events. Percutaneous PFO closure was offered to all patients with ischemic cerebrovascular event of unknown cause and presumed paradoxical embolism. **Results:** Eighteen patients aged between 2 and 17 years (median 11.5 years) were included in the study: 12 patients suffered ischemic stroke and six with transient ischemic attack (TIA). In six patients, ischemic stroke was of unknown cause and contrast TCD with VM was positive in four of them. In addition, TCD study was positive in five patients evaluated because of TIA. Nine patients with presumed paradoxical embolism underwent an attempt at the percutaneous PFO closure. **Conclusions:** It appears that the role of PFO in ischemic cerebrovascular event of unknown cause in children may be underestimated. Contrast TCD with VM is a sensitive, noninvasive method for PFO detection, proved in our experience particularly suitable for children. In children with unexplained ischemic cerebrovascular event and presumed paradoxical embolism, percutaneous PFO closure should be considered.

Key words: stroke; transcranial Doppler; closure; Amplatzer

INTRODUCTION

Ischemic cerebrovascular events are increasingly diagnosed in children and when they occur, serious consequences may result [1,2]. Despite extensive diagnostic evaluation, the cause of ischemic cerebrovascular event remains undetermined in a high percentage of affected children.

In adults, studies have demonstrated a significantly higher prevalence of patent foramen ovale (PFO) in patients with unexplained ischemic stroke than in normal controls [3,4]. Paradoxical embolism across the PFO is the presumed mechanism of stroke in some of these patients. In contrast, data regarding the possibility of paradoxical embolism in unexplained ischemic cerebrovascular event in children are sparse [5,6].

In adults, contrast transesophageal echocardiography (TEE) is the standard method for diagnosis of PFO, especially when performed with Valsalva maneuver (VM) [7]. Contrast transcranial Doppler (TCD) performed with VM proved in adults a comparably sensitive method for detection of a right-to-left shunt [8]. In

University Medical Center, Ljubljana, Slovenia

*Correspondence to: Tomaž Podnar, MD, PhD, Department of Pediatric Cardiology, University Medical Center, Vrazov trg 1, Ljubljana 1525, Slovenia. E-mail: tomaz.podnar@mf.uni-lj.si

Received 18 April 2007; Revision accepted 8 June 2007

DOI 10.1002/ccd.21305
Published online in Wiley InterScience (www.interscience.wiley.com).

contrast to the TEE, the TCD study is noninvasive. However, its use for detection of a right-to-left shunt across the PFO in children was not reported so far.

Percutaneous PFO closure is being increasingly performed for secondary prevention in adult patients with unexplained ischemic cerebrovascular events [9]. In contrast, PFO closure is performed only exceptionally in children [6].

The aims of the present study were: (1) to consider the possible role of PFO in ischemic cerebrovascular event of unknown cause in children; (2) to clarify the role of contrast TCD with VM for evaluation of children with unexplained ischemic cerebrovascular event; (3) to suggest on approach particularly suitable for the management of children with unexplained ischemic cerebrovascular event and presumed paradoxical embolism.

MATERIALS AND METHODS

All consecutive children evaluated at the Department of Pediatric Neurology, University Medical Center Ljubljana, Slovenia from January 2005 to March 2007, due to ischemic stroke or transient ischemic attack (TIA) were included in the present study. Since January 2005, all children with unexplained ischemic cerebrovascular event underwent contrast TCD with VM. This was a retrospective study. The Department of Pediatric Neurology is a single pediatric neurology unit in Slovenia and all children who suffer cerebrovascular event in the country are referred to the Department. The diagnosis of ischemic stroke based on focal neurological deficit of sudden onset, lasting more than 24 hr and the corresponding findings on magnetic resonance imaging/magnetic resonance angiography (MRI/MRA) or contrast angiography [10]. A TIA was defined as a focal neurological deficit of sudden onset, resolving completely within 24 hr. Patients with mimics of childhood stroke were excluded from the study [11].

The study was approved by an authorized ethics committee.

Study Protocol

All patients were evaluated following a protocol presented in Table I. In addition, a contrast TCD with VM was performed in patients with unexplained ischemic cerebrovascular event. Aspirin (3–5 mg/kg body weight) was given in all patients with unexplained event [10,12]. Percutaneous PFO closure was offered to all patients with unexplained ischemic cerebrovascular event and presumed paradoxical embolism.

TABLE I. Protocol for Evaluation of Children with Cerebrovascular Event

1. Neuroimaging
2. Electroencephalography
3. Haematology
Complete blood count and differential, C-reactive protein
Clotting profile
Activated protein C resistance
Factor V Leiden gene mutation
prothrombin gene mutation
Protein C, protein S
Antithrombin III
Factors VIII and XII, von Willebrand factor
Heparin cofactor II
plasminogen
Lupus anticoagulants, anticardiolipin antibodies
4. Biochemistry
Plasma and cerebrospinal fluid lactate
Plasma ammonia
Plasma urate
Plasma amino acids
Urine organic acids
Plasma homocysteine and thermolabile MTHFR gene mutation
Lipid status (lipoprotein A, fasting cholesterol)
Alpha -1- antitripsin
5. Microbiology: Borrelia, Mycoplasma, and viral serology and appropriate additional infection screen of serum and cerebrospinal fluid, depending on the clinical picture
6. Electrocardiography
7. Transthoracic echocardiography
8. Contrast TCD with Valsava maneuver (in a patient with an unexplained event)

MTHFR, methylene tetrahydrofolate reductase; TCD, transcranial Doppler.

Neuroimaging Protocol

Neuroimaging was performed according to the previously published protocol [13]. All children underwent brain MRI/MRA at diagnosis and during follow-up. At diagnosis, contrast angiography was performed in all patients with stroke. MR images were obtained with 1,5T MR tomograph (SIGNA Horizon; General Electric Medical System, Milwaukee, WI). MRI studies included T1-weighted, T2-weighted and fluid-attenuated inversion recovery (FLAIR) sequences, diffusion weighted imaging (DWI), and three-dimensional time-of-flight (TOF) MRA. Conventional contrast cerebral angiography studies included 4-vessel angiography.

Contrast TCD Protocol

All patients with unexplained ischemic cerebrovascular event underwent contrast TCD with VM to detect a right-to-left shunt [14]. During the examination, patients were in a supine position and both middle cerebral arteries were monitored. All TCD examinations were performed with a Multi-Dop X4 (DWL, Sipplingen, Germany). The arterial velocity was recorded in the left and right middle cerebral artery through temporal acoustic windows using 2 MHz transducers. Microcavitation saline contrast was generated by mixing 9 mL of 6% hydroxyethyl starch in isotonic sodium chloride solution (HAES-steril 6%) and 1 mL of air between two 10-mL syringes connected to a three-way stopcock, which was attached to an intravenous catheter inserted into a cubital vein [15]. Once the contrast was prepared, it was immediately injected as a bolus before a VM which lasted 10 sec. During VM patients maintained pressure of 20 mm Hg. Special software for emboli count was used to record and count high-intensity transient signals from each middle cerebral artery. The appearance of microbubbles in the middle cerebral arteries was automatically recorded by the equipment together with the time, side, and intensity (in decibels) of each event. The test was considered positive when at least one microbubble was detected within 22 sec of the contrast injection. When the result was negative or questionable, the test was repeated.

Percutaneous PFO Closure Protocol

Decision to offer percutaneous PFO closure was made jointly by a pediatric neurologist and pediatric cardiologist. An informed written consent was obtained from all parents or guardians of the children. At the time of the procedure, all patients were on aspirin (3–5 mg/kg body weight). Intravenous antibiotic prophylaxis (cephazoline) was administered in all patients. All procedures were performed under general endotracheal anesthesia. Initially, a precise TEE examination of the interatrial septum was performed using two-dimensional and color Doppler technique. Contrast TEE was not routinely performed. Implantation procedure was guided both by fluoroscopy and TEE. A 6F sheath was placed in the right femoral vein. The PFO was crossed using a 5F multipurpose catheter. Heparin (100 IU/kg body weight) was administered intravenously. The tip of the catheter was positioned into the left upper pulmonary vein and exchanged with an 8F long delivery sheath. An Amplatzer PFO occluder (AGA Medical, Golden Valley, MN) was implanted in all patients [16]. A detailed TEE examination was repeated after the device was released. Dalteparin sodium (50 U/kg body weight) was administered subcutaneously after the procedure and 12 hr later. Infective endocarditis prophylaxis and aspirin (3–5 mg/kg body weight) were recommended for 6 months after the procedure.

Follow-up

All patients were evaluated at intervals of 6 months by a pediatric neurologist. They underwent follow-up MRI/MRA 6 months after the cerebrovascular event.

After the percutaneous PFO closure, electrocardiogram, chest radiograph, and transthoracic echocardiogram (TTE) were performed 24-hr after the procedure. Thereafter, follow-up electrocardiogram and TTE were performed at 1 month, 3 months, and 1 year. Contrast TCD with VM was repeated 6 months after the implantation procedure.

Statistical Analysis

The data are expressed as median and ranges.

RESULTS

Eighteen patients aged between 2 and 17 years (median, 11.5 years) were included in the present study. There were nine girls and nine boys.

Twelve patients suffered an ischemic stroke (Tables II and III). In six patients, the stroke was of unknown cause and contrast TCD study was positive in four of them. The patient GP suffered right sided hemiparesis. A week later she suffered three TIAs with left sided hemiparesis and paresthesias. MRI revealed infarction in the left basal ganglia. Stenosis of the left middle cerebral artery and left anterior cerebral artery were delineated by contrast angiography. A contrast TCD with VM was performed and was positive. She underwent successful percutaneous PFO closure and no further cerebrovascular events were noticed afterwards. A 17-year-old patient KS was taking oral contraceptives at the time of the stroke. Patient GT was heterozygous for methylene tetrahydrofolate reductase gene mutation. Percutaneous PFO closure was successfully performed in all four patients with presumed paradoxical embolism.

In six remaining patients with ischemic stroke, the cause of the stroke was clearly established. The patient KD underwent surgical closure of sinus venosus-type atrial septal defect at the age of 3 years. He suffered ischemic stroke at the age of 7 years and thereafter, MRA revealed multiple stenoses of cerebral arteries. Negative contrast TCD with VM excluded residual shunt across the interatrial septum.

Six patients in the study suffered TIA (Table IV). Results of neuroimaging evaluation (MRI/MRA) were normal in all of them. In three patients, congenital hypercoagulable state was detected and contrast TCD was positive in all of them. Altogether, contrast TCD with VM was positive in five patients after TIA and they all underwent an attempt at the percutaneous PFO

TABLE II. Neuroimaging Findings in Stroke Patients

No.	Patients	MRI findings	MRA	CA
1.	KD	Infarction in the left cerebellar hemisphere	Occlusion of the left PICA	Multiple stenoses of vertebral arteries and severe stenosis of left PICA
2.	SD	Subcortical infarction in the left frontal lobe	Normal	Normal
3.	GP	Infarction in the left basal ganglia	Possible stenosis of the left MCA	Stenosis of the left MCA and left ACA-A1
4.	SH	Infarction in the ventral part of right thalamus and internal capsule	Possible aneurysm of the ACA right sided	Aneurysm of the ACA right sided
5.	PM	Infarction in the left frontal and temporal lobe	Stenoses of the left MCA, distal part of ICA and persistent left trigeminal artery	Stenoses of the left MCA, distal part of ICA and persistent left rigeminal artery
6.	KS	Two small subcortical infarctions in the right frontal and parietal lobe	Normal	Normal
7.	GT	Infarction in the left putamen	Normal	Normal
8.	MA	Infarction of the posterior part of external capsule and lentiform nucleus and the middle part of corona radiata	Stenoses of ICA, MCA occluded	Stenoses of the left ICA, occlusion of the MCA and stenoses of left ACA
9.	SM	Infarction in the right corpus striatum and capsula interna	Normal	Normal
10.	HJ	Large infarction in the left MCA territory	Occlusion of the left ICA and left MCA	Occlusion of the left ICA
11.	EAS	Infarction in the left basal ganglia	Normal	Normal
12.	KU	Large infarction in the right frontal lobe (cortical and subcortical) and in the right basal ganglia and internal capsule	Multiple stenoses of the right ICA and ACA, right MCA occluded	Multiple stenoses of the right ICA and ACA, right MCA occluded

ACA, anterior cerebral artery; CA, contrast angiography; ICA, internal carotid artery; MCA, middle cerebral artery; MRI/MRA, magnetic resonance imaging/angiography; PICA, posterior inferior cerebellar artery.

closure. In patient PP, despite multiple attempts, we were not able to cross interatrial septum. A right-to-left shunt across the interatrial septum was not seen by color Doppler TEE. Contrast TEE, performed by agitated plasma expander injected through the catheter into the tunnel of the foramen ovale failed to demonstrate a right-to-left shunt across the PFO. In addition, pulmonary arteriography was performed during cardiac catheterization to exclude pulmonary arteriovenous malformation as a possible cause of paradoxical embolism. Percutaneous PFO closure was successfully performed in the remaining four patients after TIA.

Transthoracic echocardiography was performed in all 18 patients in the study and excluded intracardiac thrombus or vegetation in all of them. All 12 patients with unexplained cerebrovascular event underwent detailed TTE evaluation of the interatrial septum using two-dimensional and color Doppler echocardiography. In seven patients, interatrial septum was assessed as normal. Evaluation of the interatrial septum was inadequate in four patients. Color Doppler TTE proved a right-to-left shunt across the interatrial septum in one patient. Her contrast TCD study was highly positive–20 microemboli.

TEE was performed in all nine patients who underwent an attempt at percutaneous PFO closure. Aneurysm of the interatrial septum was excluded in all of them. In five patients, color Doppler TEE demonstrated a right-to-left shunt across the PFO. In the remaining four patients, we were not able to prove a right-to-left shunt across the interatrial septum using color Doppler TEE.

No complications or cerebrovascular events were detected in a group of eight patients after percutaneous PFO closure during the follow-up period ranging from 6 to 21 months (median 15). All eight patients were $\geq$6 months after implantation procedure and contrast TCD with VM was negative in seven of them. In a patient GT, who suffered ischemic stroke, 5 microemboli were detected at 6 months follow-up TCD evaluation.

DISCUSSION

According to the results of the present study, it appears that the role of PFO in ischemic cerebrovascular event of unknown cause in children may be underestimated. Contrast TCD with VM is a sensitive, noninvasive method for PFO detection, proved in our

TABLE III. Summary of Stroke Patients

No.	Patients	Age [years]/sex	Clinical presentation	Diagnosis	Probable cause	Associated condition	TCD	PFO closure
1.	KD	7/M	Vertigo, vomitus ataxia, R hemiparesis	IS + TIA	Multiple stenoses of brain vessels	Multiple stenoses: pulmonary vessels, left renal artery, CoA–stent, MTHFR HeZ	NEG	/
2.	SD	12/M	R hemiparesis and paresthesias, aphasia	IS unknown cause	PE	/	2 MES	Closed
3.	GP	14/F	1. R hemiparesis (IS) 2. L hemiparesis and paresthesias (TIA)	IS unknown cause + 3 TIAs	Angiopathy or PE	/	1 MES	Closed
4.	SH	14/F	L hemiparesis, headache, vomiting	IS, meningo-encephalitis	Borrelia Burgdorferi infection	Aneurysm of anterior communicating artery-right sided	/	/
5.	PM	15/M	R hemiparesis, aphasia	IS	Primary CNS vasculitis	/	/	/
6.	KS	17/F	R hemiparesis	IS unknown cause	PE	Oral contraceptives	20 MES	Closed
7.	GT	9/F	R hemiparesis	IS unknown cause	PE	MTHFR HeZ	11 MES	Closed
8.	8 MA	16/M	R hemiparesis, dysphasia, visual disturbances	IS	Primary CNS vasculitis	/	/	/
9.	SM	5/F	L hemiparesis	IS unknown cause	/	/	NEG	/
10.	HJ	2/M	R hemiparesis	IS	Postvaricella angiopathy	/	/	/
11.	EAS	10/F	R hemiparesis	IS unknown cause	/	/	NEG	/
12.	KU	10/F	L hemiparesis and paresthesias	IS	Primary CNS vasculitis	MTHFR HeZ	/	/

ASD, atrial septal defect; CNS, central nervous system; CoA, coarctation of aorta; F, female; HeZ, heterozygote; HoZ, homozygote; IS, ischemic stroke; L, left sided; M, male; MES, microembolic signal; MTHFR, methylene tetrahydrofolate reductase; NEG, negative result; PE, paradoxical embolism; PFO, patent foramen ovale; R, right sided; TIA, transient ischemic attack.

TABLE IV. Summary of Transient Ischemic Attack Patients

No.	Patients	Age [years]/sex	Clinical presentation	Duration of signs/symptoms [hr]	Probable cause	Associated conditions	EEG	TCD	PFO closure
1.	HM	9/F	R hemiparesis, 3 hr later short headache, aphasia	7, 5	Unknown cause	/	Discharges of generalized spike-waves, no clinical signs	NEG	/
2.	KA	11/M	R hemiparesis, dysphasia, mild headache 24 hr	22	PE	Factor V Leiden and MTHFR HeZ	Slower activity over left centro-temporal region; follow-up normal	13 MES	Closed
3.	KK	14/F	L hemiparesis, mild headache 2 hr	10	PE	/	Normal	27 MES	Closed
4.	PP	11/M	R hemiparesis, aphasia, transitory amnesia	4	PE	/	Normal	3 MES	Attempt
5.	PD	17/M	Aphasia (motoric and sensoric)	5	PE	MTHFR HeZ	Normal	19 MES	Closed
6.	BN	13/M	R hemiparesis with mild headache	22	PE	MTHFR HoZ	Slower activity over right temporal region, follow-up normal	6 MES	Closed

EEG, electroencephalography; F, female; HeZ, heterozygote; HoZ, homozygote; L, left sided; M, male; MES, microembolic signal; MTHFR, methylene tetrahydrofolate reductase; NEG, negative result; PE, paradoxical embolism; PFO, patent foramen ovale; R, right sided.

experience particularly suitable for children. In all children with unexplained ischemic cerebrovascular event and presumed paradoxical embolism, percutaneous PFO closure should be considered.

Our study was limited by a small number of involved patients reflecting relative rarity of ischemic cerebrovascular events in children [1,2]. Patient population in the study was highly representative as our Department of Pediatric Neurology is a single pediatric neurology unit in the country and all children with cerebrovascular events are referred to the Department. Despite rarity, serious consequences may occur in affected patients, additionally compounded by their young age. Therefore, children are extensively evaluated after ischemic cerebrovascular event to determine the cause and if possible, to prevent recurrent events. Despite that, the cause of ischemic stroke remains unexplained in a high percentage of affected children [1,2]. Furthermore, the risk of recurrent childhood stroke ranges from 20 to 40% [1,2].

In adults, studies have demonstrated a significantly higher prevalence of PFO in patients with unexplained stroke than in normal controls [3,4]. Paradoxical embolism across the PFO is the supposed mechanism of stroke in some of these patients. In contrast, data regarding presumed paradoxical emboli in childhood ischemic cerebrovascular event of unknown cause are sparse. PFO recognition and percutaneous PFO closure in children and young adults was the major point of the two recent reports. Agnetti et al. reported two children with unexplained stroke because of a presumed paradoxical embolism across the PFO [5]. Children were also involved in a study regarding percutaneous atrial septal defect and PFO closure for paradoxical emboli in children and young adults [6]. Only patients selected for percutaneous closure were included in the study. In addition, the median age of patients in that study was 29.0 years and the number of children involved was not mentioned. In the present study, 12 patients suffered ischemic stroke and in six of them the stroke remained unexplained. In four of six patients without clearly established cause of stroke, a contrast TCD with VM was positive suggesting paradoxical embolism across the PFO as a possible cause of ischemic stroke. TCD was not offered to all 18 patients in this study because we believe that performing TCD in a patient with an obvious explanation for cerebrovascular event would only create confusion because of the high prevalence of PFO in population. Prothrombotic abnormalities have been identified in 20–50% of children presenting with ischemic stroke [17]. In addition, venous thromboembolism is being diagnosed more frequently in children [18]. Both congenital and acquired conditions contribute to the development of venous thrombosis in children. The highest incidence is during the neonatal period, followed by another peak in adolescence. As childhood stroke is defined as a cerebrovascular event between 30 days and 18 years of age, neonates were not included in our study [1]. In two of our four patients with unexplained ischemic stroke and positive contrast TCD study, a known prothrombotic condition was detected: oral contraceptives in one patient and heterozygosity for MTHFR mutation in another patient. As in most adult patients, paradoxical embolism could not be proved in our patients. However, detection of a known prothrombotic condition further increases likelihood of paradoxical embolism as a cause of ischemic stroke. Furthermore, multiple risk factors are found in many ischemic strokes in children and may predict stroke recurrence [19]. All four patients with unexplained stroke and positive contrast TCD in our study underwent successful percutaneous PFO closure. We performed percutaneous PFO closure also in a patient (patient GP, Table III) with ischemic stroke despite angiopathy detected by contrast angiography. This patient initially suffered right sided hemiparesis. Neuroimaging revealed infarction in the left basal ganglia and stenosis of the left middle cerebral artery and left anterior cerebral artery. One week later, she suffered three TIAs with left sided hemiparesis and paresthesias. No further cerebrovascular events were detected in this patient after percutaneous PFO closure. Among six patients who suffered TIA, contrast TCD study was positive in five patients and known prothrombotic abnormalities were detected in three of them. All five patients with TIA and positive TCD study underwent percutaneous PFO closure and procedure was successfully performed in four of them. Prothrombotic conditions detected in this study were relatively mild, and therefore Warfarin was not given after percutaneous PFO closure in any of our patients.

On the basis of data derived from adult patient population it is often assumed that children lack the potential for venous thrombosis and thus, in contrast to adult patients, paradoxical embolism is often less stringently considered as a possible cause of unexplained childhood ischemic stroke [20,21]. Therefore, children who suffered unexplained ischemic cerebrovascular event are frequently not adequately evaluated for PFO. In adult patient population, contrast TEE is the standard method for PFO detection, especially when performed with VM [7]. Contrast and color flow Doppler transthoracic echocardiography (TTE) is noninvasive and is a possible alternative method for children [6]. However, sensitivity and specificity of contrast TTE for PFO detection in adult patients is low and similar data in children are lacking [22]. Despite that, most pediatric centers rely on contrast TTE for PFO detection [2]. If the TTE is unrevealing TEE is usually performed. However, deep sedation is necessary for adequate TEE in children thus preventing VM to be performed during examination [23]. In comparison, contrast TCD with VM proved as a highly sensitive and accurate method for detection of right-to-left shunts in adults [8]. The method is noninvasive and performed without sedation making it particularly suitable for children. Pressure maintained during VM is measured thus ensuring standard test conditions. In addition, contrast TCD allows quantification of the right-to-left shunt by counting the number of detected bubbles [24]. So far, data regarding use of contrast TCD with VM and quantification of the right-to-left shunt in pediatric patient population are lacking. In the present study, a contrast TCD with VM was attempted in 13 patients and was successfully performed in all of them, the youngest being 5-years old. Pressure measurement during VM confirmed that the study was adequately performed in all of them. However, small children are not able to perform adequate VM and would not be suitable for contrast TCD with VM. Therefore, we would have to rely on TTE for PFO detection in small children. If thrombus or vegetation cannot be excluded by TTE in any of our patients we would then perform TEE. In the present study, PFO was actually crossed by catheter in all but one patient during subsequent cardiac catheterization. Three microembolic signals were detected by contrast TCD study in this patient and we speculate that occasional microemboli may cross normal pulmonary vascular bed.

The secondary prevention in adult patients after unexplained ischemic cerebrovascular event is controversial [25]. Percutaneous PFO closure is increasingly being performed in adult patients and evidence suggests that it is a safe and promising technique in patients with unexplained ischemic cerebrovascular event [9]. The secondary prevention in children with unexplained ischemic cerebrovascular event is based on extrapolation of data from the adult literature to children [1,2]. Therefore, aspirin at a dose 3–5 mg/kg is mostly used in children. However, the recurrence rate in childhood ischemic cerebrovascular event is up to 40% [1,2]. Percutaneous PFO closure seems particularly attractive option for secondary prevention in children with unexplained ischemic cerebrovascular event and therefore, we performed procedure in this group of patients. However, before embarking on cardiac catheterization and percutaneous PFO closure every effort should be made to determine the cause of cerebrovascular event in each particular child. In this way, only children with a high probability of paradoxical embolism will be subjected to the minimal risk of percutaneous PFO closure and will thereafter enjoy a lifelong benefit of the procedure.

In adults, contrast TEE with VM is usually performed to detect a right-to-left shunt across the PFO. At the same time, detailed interrogation of the interatrial septum is performed to detect associated aneurysm of the interatrial septum or atrial septal defect as a possible route for paradoxical embolism. In addition, trombus and vegetation are sought by TEE. Later, percutaneous PFO closure may be performed safely even in a wake patient using solely fluoroscopic guidance. In children, deep sedation is necessary for satisfactory TEE examination thus preventing appropriate VM during contrast study. Therefore, in all children with positive contrast TCD study and unexplained ischemic cerebrovascular event we attempted percutaneous PFO closure without previous TEE evaluation. TEE was performed for the first time in the catheterization laboratory, under general endotracheal anesthesia. Immediately after TEE evaluation, we proceeded to the percutaneous closure using both TEE and fluoroscopic guidance.

Six months after the percutaneous PFO closure, contrast TCD with VM proved residual right-to-left shunt in one patient. After device implantation, her shunt diminished from 11 to 5 MES. According to the reported experience, we can expect complete PFO closure during further follow-up [9].

CONCLUSION

Our data suggest that paradoxical embolism across the PFO may be an underestimated cause of unexplained ischemic cerebrovascular event in children. Contrast TCD with VM is particularly suitable for PFO detection in children and should be performed in all children with unexplained ischemic cerebrovascular event. Percutaneous PFO closure should seriously be taken into consideration in all children with unexplained ischemic cerebrovascular event and presumed paradoxical embolism. However, further studies including a larger number of patients and long-term follow-up are necessary to support our conclusions.

REFERENCES

1. Lynch JK, Hirtz DG, DeVeber G, Nelson KB. Report of the National Institute of Neurological Disorders and Stroke workshop on perinatal and childhood stroke. Pediatrics 2002;109: 116–123.
2. Jordan LC. Stroke in childhood. Neurologist 2006;12:94–102.
3. Lechat P, Mas JL, Lascault G, Loron P, Theard M, Klimczac M, Drobinski G, Thomas D, Grosgogeat Y. Prevalence of patent foramen ovale in patients with stroke. N Engl J Med 1988;318:1148–1152.
4. Webster MW, Chancellor AM, Smith HJ, Swift DL, Sharpe DN, Bass NM, Glasgow GL. Patent foramen ovale in young stroke patients. Lancet 1998;2:11–12.
5. Agnetti A, Carano N, Sani E, Tchana B, Allegri V, Bernasconi S, Squarcia U. Cryptogenic stroke in children: Possible role of patent foramen ovale. Neuropediatrics 2006;37:53–56.
6. Bartz PJ, Cetta F, Cabalka AK, Reeder GS, Squarcia U, Agnetti A, Aurier E, Carano N, Tachana B, Hagler DJ. Paradoxical emboli in children and young adults: Role of atrial septal defect and patent foramen ovale device closure. Mayo Clin Proc 2006;81:615–618.
7. Chen WJ, Kuan P, Lien WP, Lin FY. Detection of patent foramen ovale by contrast transesophageal echocardiography. Chest 1992;101:1515–1520.
8. Job FP, Ringelstein EB, Grafen Y. Comparison of transcranial contrast Doppler sonography and transesophageal contrast echocardiography for the detection of patent ovale in young stroke patients. Am J Cardiol 1994;74:381–384.
9. Braun M, Gliech V, Boscheri A, Schoen S, Gahn G, Reichmann H, Haass M, Schraeder R, Strasser RH. Transcatheter closure of patent foramen ovale (PFO) in patients with paradoxical embolism. Periprocedural safety and mid-term follow-up of three different device occluder systems. Eur Heart J 2004;25:424–430.
10. Pediatric Stroke Working Group. Stroke in childhood: Clinical guidelines for diagnosis, management and rehabilitation, 2004. Available at: http://www.rcplondon.ac.uk/pubs/books/childstroke.
11. Shellhaas RA, Smith SE, O'Tool E, Licht DJ, Ichord RN. Mimics of childhood stroke: Characteristics of a prospective cohort. Pediatrics 2006;118:704–709.
12. Monagle P, Chan A, Massicotte P, Chalmers E, Michelson AD. Antithrombotic therapy in children: The seventh ACCP conference on antithrombotic and thrombolytic therapy. Chest 2004; 126:645–687.
13. Husson B, Lasjaunias P. Mini-symposium: Stroke in children. Radiological approach to disorders of arterial brain vessels associated with childhood arterial stroke—A comparison between MRA and contrast angiography. Pediatr Radiol 2004;34:10–15.
14. Devuyst G, Despland PA, Bogousslavsky J, Jeanrenaud X. Complementarity of contrast transcranial Doppler and contrast transesophageal echocardiography for the detection of patent foramen ovale in stroke patients. Eur Neurol 1997;38:21–25.
15. Seiler C. How should we assess patent foramen ovale? Heart 2004;90:1245–1247.
16. Han YM, Gu X, Titus JL, Rickers C, Bass JL, Urness M, Amplatz K. New self-expanding patent foramen occlusion device. Catheter Cardiovasc Interv 1999;47:370–376.
17. Barnes C, DeVeber G. Prothrombotic abnormalities in childhood ischaemic stroke. Thromb Res 2006;118:67–74.
18. Parasuraman S, Goldhaber SZ. Venous thromboembolism in children. Circulation 2006;113:e12–e16.
19. Lanthier S, Carmant L, David M, Larbrisseau A, de Veber G. Stroke in children. The coexistence of multiple risk factors predicts poor outcome. Neurology 2000;54:371–378.
20. Anderson FA Jr, Wheeler HB, Goldberg RJ, Hosmer DW, Patwardhan NA, Jovanovic B, Forcier A, Dalen JE. A population-based perspective of the hospital incidence and case-fatality

rates of deep vein thrombosis and pulmonary embolism. The Worcester DVT Study. Arch Intern Med 1991;151:933–938.
21. Meier B. PFO closure, when the going gets tough. Catheter Cardiovasc Interv 2005;64:345–347.
22. Belkin RN, Pollack BD, Ruggiero ML, Alas LL, Tatini U. Comparison of transesophageal and transthoracic echocardiography with contrast and color Doppler in the detection of patent foramen ovale. Am Heart J 1994;128:520–525.
23. Krauss B, Green SM. Procedural sedation and analgesia in children. Lancet 2006;367:766–780.
24. Belvis R, Leta RG, Marti-Fabregas J, Cocho D, Carreras F, Pons-Llado G, Marti-Vilalta JL. Almost perfect concordance between simultaneous transcranial Doppler and transesophageal echocardiography in the quantification of right-to-left shunts. J Neuroimaging 2006;16:133–138.
25. Sievert H, Taaffe M. Patent foramen ovale: The jury is still out. Eur Heart J 2004;25:361–362.